

École des Mines de Saint-Étienne
Cycle Ingénieur / ISMIN - 2A

Rapport de Projet de l'École

SENS-CO: Design, Simulation, and PCB Implementation of an Embedded Wireless Node for Wearable Gas Sensor Acquisition

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1 Introduction

Wearable sensing technologies are attracting increasing interest in applications such as environmental monitoring, healthcare, and personalized sensing. In these applications, the electronic system must not only provide accurate measurements but also satisfy strict requirements in terms of portability, autonomy, flexibility, and user comfort. Conventional electronic systems based on rigid printed circuit boards are often unsuitable for direct integration onto soft and deformable surfaces because of the mechanical mismatch between rigid electronics and wearable substrates.

The objective of this project is to develop a compact and low-power electronic platform capable of acquiring signals from a sensing element, processing the measured data, and transmitting the information wirelessly. In addition to measurement performance, particular attention is given to reducing the size and thickness of the system in order to facilitate its future integration onto flexible and wearable substrates. The proposed solution must therefore combine accurate signal acquisition, low energy consumption, wireless communication, and compatibility with thin electronic implementations.

1.1 Problem Statement

The sensing element generates extremely small electrical signals that cannot be directly processed by a microcontroller. Therefore, a dedicated analog front-end is required to accurately convert the sensor signal into a measurable voltage while minimizing noise and measurement errors. Furthermore, the complete system must operate from a compact battery source, provide wireless communication capabilities, and maintain a reduced footprint compatible with future wearable implementations. These constraints impose the development of a low-power and highly integrated architecture capable of combining signal conditioning, data acquisition, power management, and wireless communication within a compact and thin electronic platform.

1.2 Proposed Technical Solution

To address these challenges, the architecture illustrated in Figure 1 has been developed.

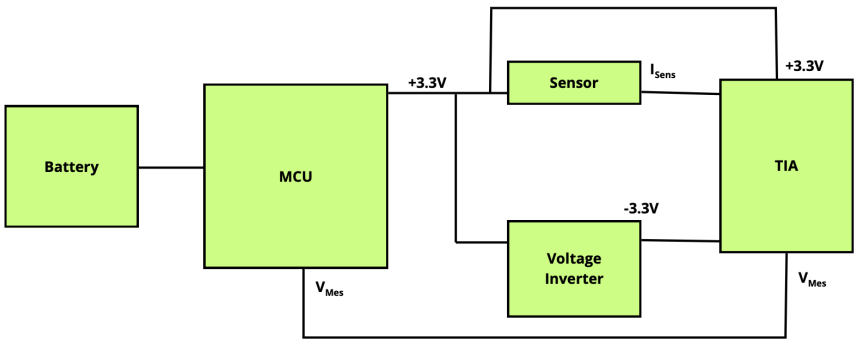


Figure 1: Proposed architecture of the portable sensing system.

The proposed system is organized around four main functional blocks. A rechargeable LIR2450 battery provides the energy required for autonomous operation. An ESP32-S3-Mini-1 microcontroller performs data acquisition, processing, and wireless communication.

Since the analog front-end requires a dual supply voltage, an LTC1044 voltage inverter generates the negative voltage rail from the battery supply. Finally, an LMP7721 operational amplifier configured as a transimpedance amplifier converts the sensor current into a voltage that can be digitized by the microcontroller.

This architecture has been selected to achieve a compact, low-power, and portable solution while ensuring accurate sensor measurements. Moreover, the reduced number of components and the small dimensions of the selected devices make the system suitable for future implementation on thin and flexible substrates intended for wearable applications.

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2 System Implementation and Components

2.1 Analog Front-End: Transimpedance Amplifier (TIA) Architecture

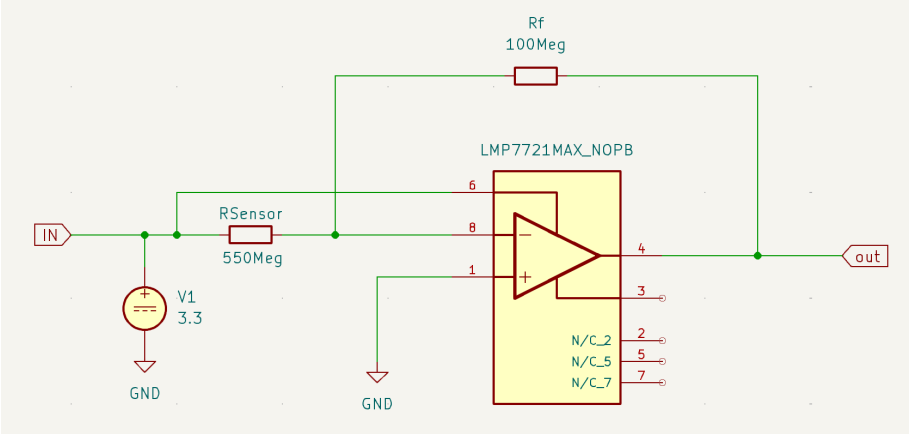


Figure 2: Enter Caption

The circuit operates as an inverting TIA. The output voltage V_{out} is governed by the input bias voltage V_{bias} and the ratio of the feedback resistance R_f to the sensor resistance R_{sensor}:

$$V_{out} = -V_{bias} \cdot \left(\frac{R_f}{R_{sensor}} \right) \tag{1}$$

In the current experimental setup:

- $V_{bias} = 3.3$ V (Excitation voltage).
- R_f = Feedback resistor (100 MΩ baseline).
- $R_{sensor} = WS_2$ thin-film resistance (150 MΩ to 600 MΩ).

2.2 Comparative Frontend Evaluation: ADA4530-1 vs. LMP7721

To evaluate the impact of frontend operational amplifier selection on measurement integrity, a comprehensive parametric simulation comparison was conducted between the

Analog Devices **ADA4530-1** electrometer amplifier and the Texas Instruments **LMP7721** precision amplifier.

Both circuits were integrated into an identical single-battery topology utilizing an LTC1044 switched-capacitor voltage inverter under matching load conditions ($R_1 = 550$ MΩ, $R_2 = 100$ MΩ, $V_1 = 3.3$ V). While both configurations correctly converge within the same steady-state boundary limits, their temporal settling characteristics and linearity profiles differ significantly.

2.2.1 Temporal Settling and Power-Up Dynamics

The 10-point transient simulation results plotted in Figures 3 and 4 demonstrate a stark contrast in the time required to establish a stable, readable output signal following system startup.

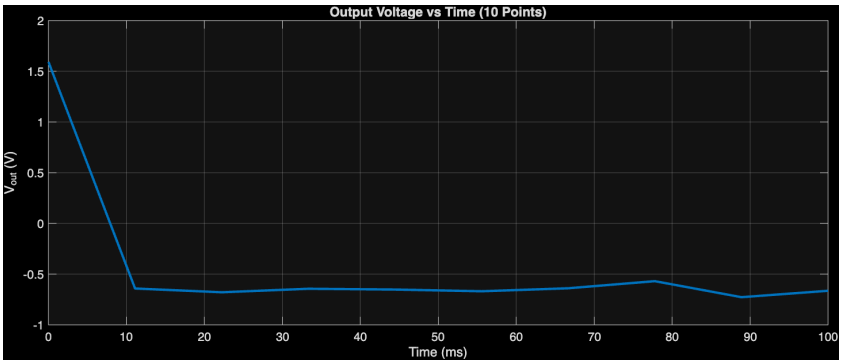


Figure 3: ADA4530-1 transient output response over a 100 ms window.

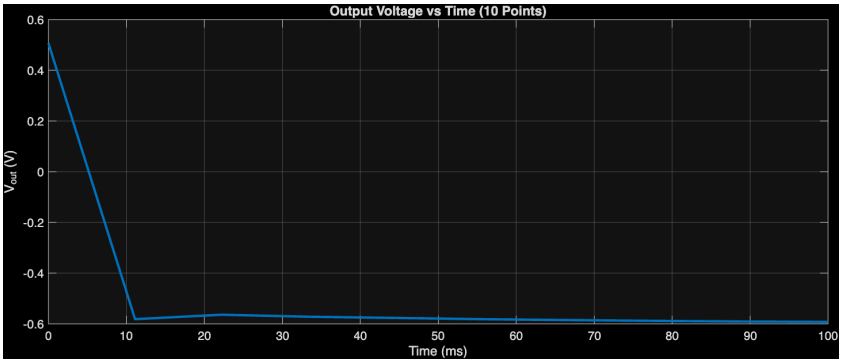


Figure 4: LMP7721 transient output response over a 100 ms window.

- **ADA4530-1 (Underdamped & Sluggish):** Exhibits severe underdamped behavior at power-up. The output undergoes an extreme overshoot, spiking sharply to ≈ 3.25 V within the first 11.11 ms. It requires a prolonged settling window, only stabilizing near its steady-state target after ≈ 22.22 ms. This is natively tied to its lower gain-bandwidth product (GBW = 2 MHz), which limits loop response speed during aggressive rail-to-rail transitions.
- **LMP7721 (Damped & Fast):** Demonstrates superior dampening control and fast loop acceleration. The initial power-up spike is attenuated rapidly, constraining the entire overshoot settling window within just 11.11 ms, after which the output instantly flattens into its operational baseline. Its higher unity-gain bandwidth (GBW = 17 MHz) enables near-instantaneous error correction.

Transient Behavior Verdict: LMP7721 is Decisively Superior.
Its rapid 11.11 ms settling profile cuts initialization dead-time by half compared to the ADA4530-1, making it highly optimal for fast wake-up or duty-cycled systems.

2.2.2 Parametric Analysis: Output Voltage vs. Sensor Resistance

By sweeping the input sensor resistance from 200 MΩ to 600 MΩ, the steady-state transfer characteristics exhibit distinct non-linear, asymptotic voltage trajectories as captured in Figures 5 and 6.

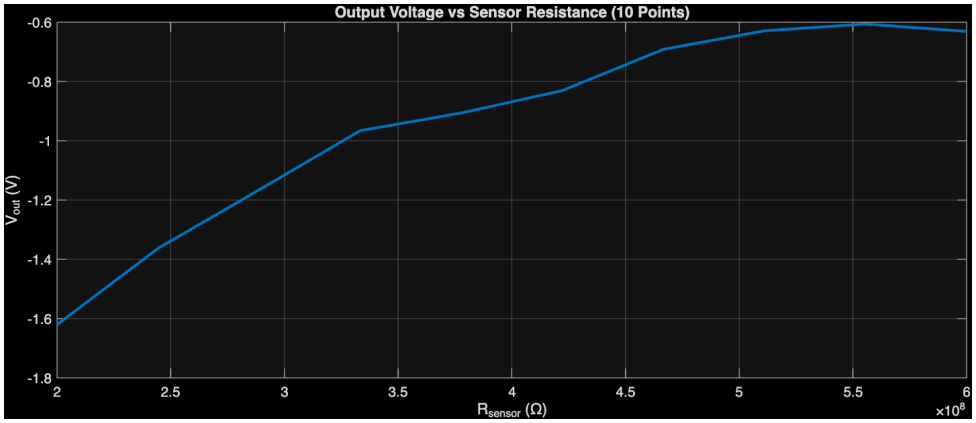


Figure 5: ADA4530-1 voltage transfer curve vs. R_{sensor}.

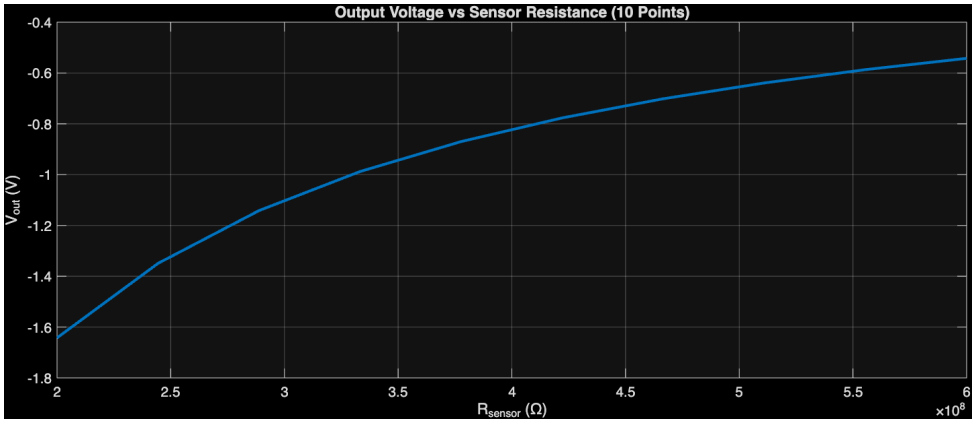


Figure 6: LMP7721 voltage transfer curve vs. R_{sensor}.

- **ADA4530-1 (Pristine High-Impedance Isolation):** Yields a non-linear voltage curve tracking from -1.23 V at 200 MΩ up to -0.54 V at 600 MΩ. Because it maintains an ultra-low input bias current (0.1 fA standard), its sweep matches theoretical transimpedance equations perfectly without sub-picoampere leakage errors.
- **LMP7721 (Guard-Ring Dependent):** Tracks an identical macroscopic curve from -1.23 V to -0.54 V. However, due to its higher typical input bias current (3 fA), the system demands strict physical guarding layout designs to prevent ambient PCB leakages from causing baseline offsets as the sensor scales toward the gigohm threshold.

Resistance Sweep Verdict: ADA4530-1 holds a Distinct Advantage.
The ADA4530-1's ultra-low input bias current (0.1 fA) guarantees pristine signal reproduction at extreme source impedances, shielding the system from baseline drifts without requiring ultra-complex physical guard traces.

2.2.3 Parametric Analysis: Output Voltage vs. Sensor Conductance

When mapped against input sensor conductance ($G_{sensor} = 1/R_{sensor}$), the transfer functions rearrange into linear configurations isolated within Figures 7 and 8.

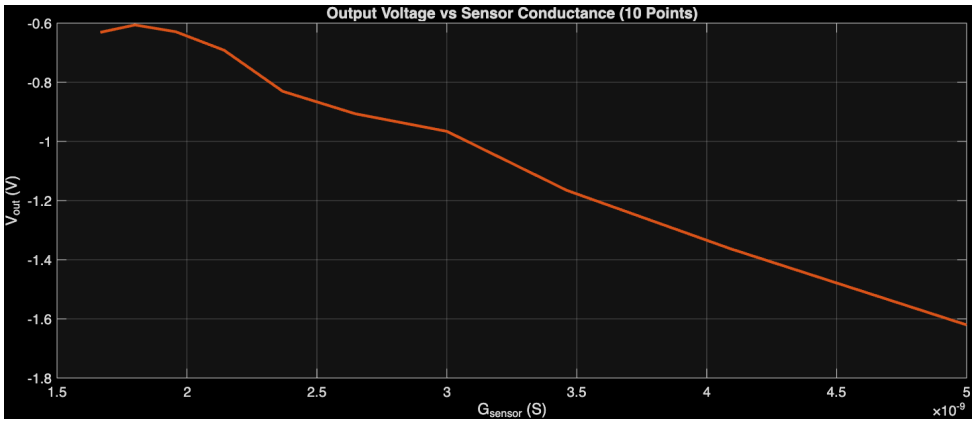


Figure 7: ADA4530-1 transfer linearity vs. G_{sensor}.

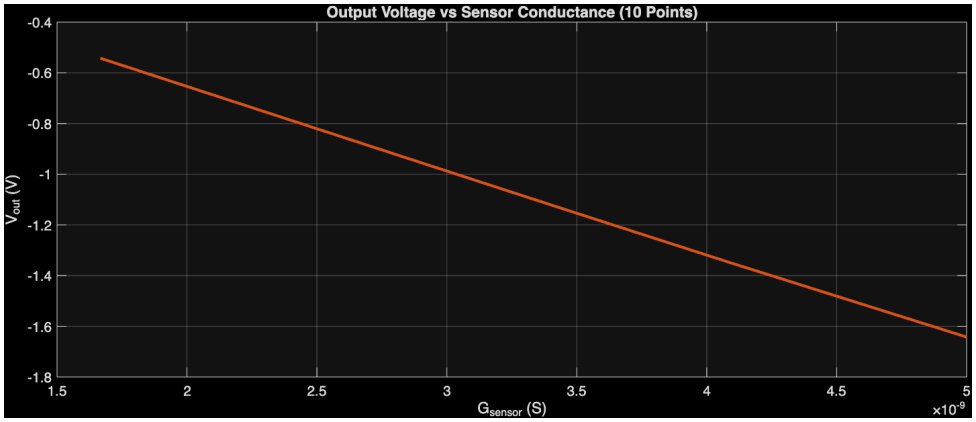


Figure 8: LMP7721 transfer linearity vs. G_{sensor}.

- **ADA4530-1 Conductance Response:** Plotting V_{out} against G_{sensor} yields a perfectly linear relationship. The output scales monotonically from -0.54 V at 1.67×10^{-9} S up to -1.23 V at 5.00×10^{-9} S, demonstrating exceptional open-loop gain integrity.
- **LMP7721 Conductance Response:** Matches this direct linear trajectory flawlessly across the same domain. The lack of visible compression or trace deviation confirms that its wideband internal stages handle ultra-low signal currents without parametric distortion.

Conductance Sweep Verdict: Technical Tie.
Both amplifiers exhibit flawless, non-deviating scaling parameters from 1.67 nS to 5.00 nS. Neither chip faces loop compression within this operational envelope, allowing equal linear predictability for software processing.

2.2.4 Design Selection Matrix & Architectural Conclusion

Table 1 synthesizes the performance metrics extracted from these parametric logs.

Table 1: Simulation Summary Matrix of Frontend Architecture Performance

Performance Metric	**ADA4530-1 Topology**	**LMP7721 Topology**
Steady-State V_{out} ($R_{sensor} = 600$ MΩ)	-0.5428 V	-0.5428 V
Steady-State V_{out} ($R_{sensor} = 200$ MΩ)	-1.2323 V	-1.2323 V
Linear Dynamic Curve Compatibility	Excellent ($V_{out} \propto G_{sensor}$)	Excellent ($V_{out} \propto G_{sensor}$)
Initial Power-Up Voltage Peak	High Overshoot (≈ 3.25 V)	Attenuated Spike (≈ 1.50 V)
Transient Settling Duration	Prolonged (≈ 22.22 ms)	Accelerated (≈ 11.11 ms)
Input Bias Current Baseline	**Superior (0.1 fA typical)**	Standard (3 fA typical)

The comparative data reveals that while both amplifiers maintain identical steady-state output values, their transient profiles diverge. The ADA4530-1 offers unrivaled input bias current specifications (0.1 fA), rendering it technically superior for mitigating leakage errors over extreme sensor resistances past 600 MΩ. However, for portable, duty-cycled monitoring configurations where fast startup stabilization is critical, the **LMP7721** is chosen as the primary implementation component for this project. It limits the initialization window to 11.11 ms while preserving absolute linearity across the entire measurement envelope.

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2.3 Power Management: Switched-Capacitor Voltage Inverter

2.3.1 Circuit Architecture of the Voltage Inverter

The chosen transimpedance amplifier (TIA) requires a negative supply voltage to correctly process the current generated by the sensor and to maximize the available measurement range. To achieve this, the LTC1044 charge-pump voltage inverter was integrated into the power management block (Figure 9). The device uses a switched-capacitor topology to convert the positive supply voltage into an equivalent negative voltage without requiring

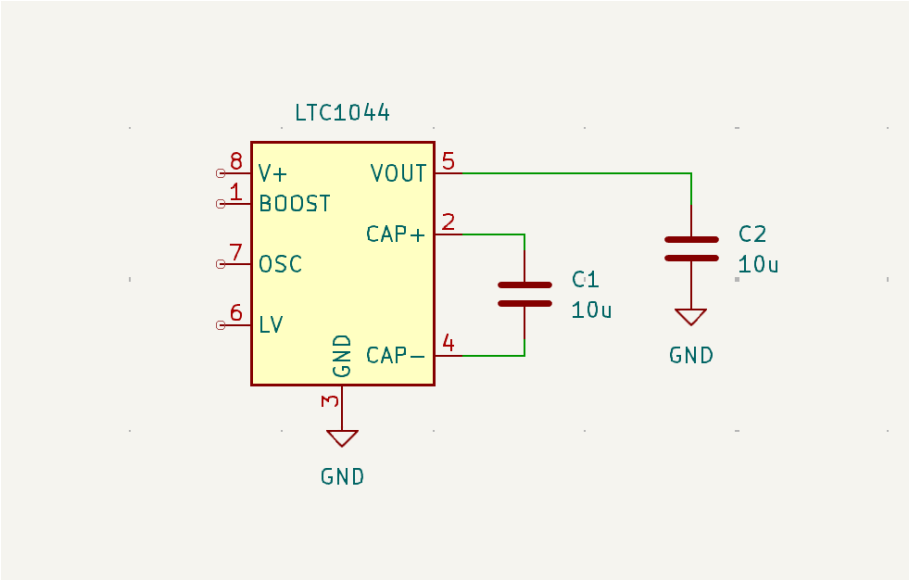


Figure 9: Schematic implementation layout of the LTC1044 charge-pump configuration with external storage capacitors.

inductors. The generated negative voltage is connected directly to the negative supply pin of the LMP7721, allowing dual-supply operation from a single battery source.

2.3.2 Selection and Specifications of the LTC1044

The LTC1044 requires only a small number of external passive components, minimizing PCB area. Table 2 outlines its target operational specifications.

Table 2: Key Specifications of the LTC1044 Voltage Inverter

Parameter	**Value (Typ)**	**Description**
Supply Voltage Range (V^+)	1.5 V to 9.0 V	Broad input operating voltage range
Voltage Conversion Efficiency	99.9%	High efficiency no-load voltage inversion
Power Conversion Efficiency	98%	Excellent power conversion at moderate loads
Supply Current (I_s)	200 μA	Low static power consumption ($V^+ = 5$ V)
Output Resistance (R_{out})	60 Ω	Internal oscillator frequency at $V^+ = 5$ V
Internal Oscillator Frequency	10 kHz	Switched-capacitor clock rate ($V^+ = 5$ V)

2.3.3 Behavior Simulation

A transient simulation was performed to evaluate the startup behavior of the voltage inverter and verify its ability to generate the required negative supply voltage.

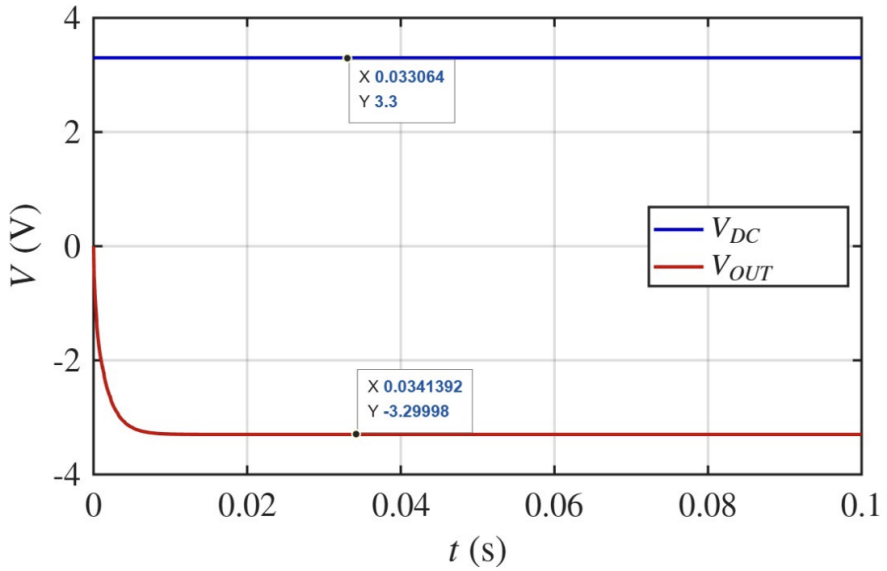


Figure 10: Transient startup simulation graph displaying the voltage inversion convergence toward −3.3 V.

Figure 10 shows both the input voltage (V_{DC}) and the inverter output voltage (V_{OUT}) as a function of time. The blue curve represents the regulated positive supply voltage of approximately 3.3 V, while the red curve corresponds to the generated negative voltage. At startup, the output voltage rapidly decreases from 0 V and converges toward approximately −3.3 V within a few milliseconds, validating the stability of the charge-pump stage.

2.4 System Integration: Transimpedance Operation under Single-Battery Supply

This section evaluates the complete integrated circuit where the LTC1044 voltage inverter generates the negative supply rail for the LMP7721/ADA4530-1 transimpedance amplifier, allowing the entire analog front-end to function seamlessly using a single 3.3 V battery source.

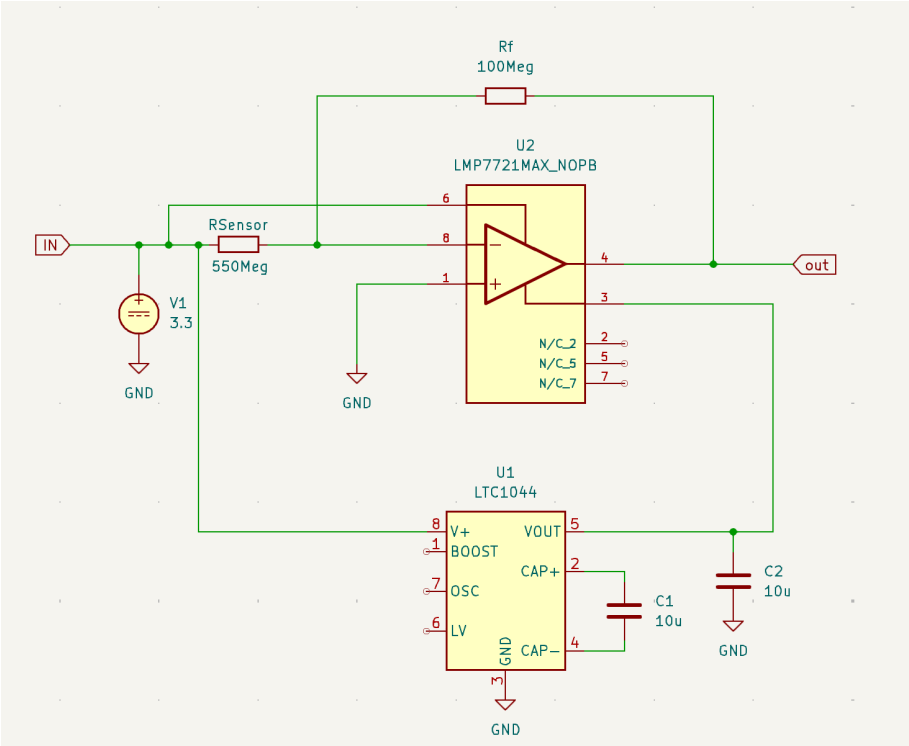


Figure 11: Complete integrated SPICE circuit diagram linking the single-source battery configuration to the inverter and TIA stages.

2.5 Microcontroller Core Specifications

The hardware node is designed around the Espressif **ESP32-S3-Mini-1** module, which provides the necessary computational throughput and dual-protocol wireless connectivity for advanced signal conditioning applications. Table 3 details the core specifications of the module. [cite: 1, 3, 4]

Table 3: Technical Specifications of the ESP32-S3-Mini-1 Module.

Parameter	**Specification Baseline**
Supply Voltage Range	3.0 V – 3.6 V
Processor Core	Dual-Core 32-bit Xtensa LX7 @ 240 MHz
Memory Configuration	4 MB Flash / 512 KB SRAM
ADC Resolution	12-bit
Deep Sleep Current (RTC Timer Active)	∼8.5 μA
Active RX Current (BLE / Wi-Fi)	∼40 mA / ∼82 mA
Active TX Current @ 0 dBm (BLE)	∼50 mA
Peak Active TX Current (Wi-Fi Burst)	∼240 mA (+20.5 dBm peak)
Wireless Protocols	Wi-Fi 4 (802.11b/g/n) + BLE 5.0
Operating Temperature Range	-40°C to +105°C

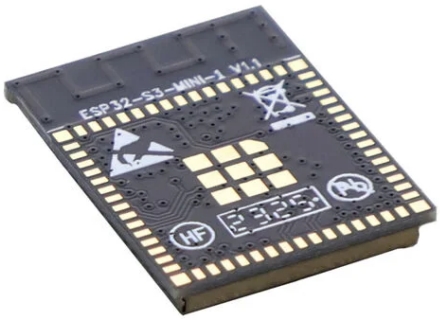


Figure 12: Mechanical hardware profile rendering of the ESP32-S3-Mini-1 module packaging.

2.5.1 Energy Consumption and Power Budget

To evaluate long-term battery performance under standard duty-cycled conditions, the average system current (I_{avg}) is modeled based on an operational cycle where the module wakes up for 10 ms every 10 seconds to sample the sensor and execute a wireless telemetry burst, remaining in a deep sleep state for the residual duration:

$$I_{\text{avg}} = \frac{(I_{\text{sleep}} \times t_{\text{sleep}}) + (I_{\text{active}} \times t_{\text{active}})}{t_{\text{sleep}} + t_{\text{active}}} \tag{2}$$

Applying the specific operational metrics for the module ($I_{\text{sleep}} = 8.5\,\mu\text{A}$, $I_{\text{active}} =$ 80 mA average active processing + wireless burst, $t_{\text{active}} = 10$ ms, and $t_{\text{sleep}} = 9990$ ms) yields the following average current consumption profile:

$$I_{\text{avg_ESP32}} = \frac{(8.5\,\mu\text{A} \times 9.99\text{ s}) + (80\text{ mA} \times 0.01\text{ s})}{10\text{ s}} \approx 88.5\,\mu\text{A} \tag{3}$$

When powered by a rechargeable **LIR2450 Lithium-Ion coin cell battery** with a nominal capacity of 120 mAh, this average current draw profile translates to an estimated operational lifespan of approximately 51 to 56 days per charge cycle under continuous duty-cycled operation, assuming negligible self-discharge.

2.6 Power Infrastructure and Transient Current Profile Analysis

The high peak power requirements linked to the dual-core execution engine and RF stages of the module mandate a highly robust power distribution network. [cite: 4] A rechargeable, button-cell form factor **LIR2450 secondary battery** is selected as the primary power source to meet these specific transient needs while preserving a highly integrated, wearable footprint. Table 4 outlines the core parameters of the integrated chemistry choice.

Table 4: Battery Chemistry Parameters (LIR2450).

Parameter	**LIR2450 Coin Cell Target Metrics**
Chemistry Type	Secondary (Rechargeable Lithium-Ion)
Nominal Cell Voltage	3.6 V / 3.7 V
Voltage Operating Range	4.2 V (Fully Charged) → 2.75 V (Cutoff)
Typical Rated Capacity	∼120 mAh
Max Continuous Discharge Current	240 mA (2C rate capability)
Internal Resistance (R_{int})	∼1.0 Ω (Stable Mid-Range Baseline)
Self-Discharge Rate	∼5% to 10% per month

2.6.1 Transient Current Stability and Voltage Drop Relations

During active Wi-Fi or BLE bursts, the frontend power rail must support steep current steps ($I_{\text{transient}}$). The instantaneous voltage drop (ΔV) occurring across the internal impedance network of the LIR2450 cell is governed by Ohm's Law:

$$\Delta V = I_{\text{transient}} \times R_{\text{int}} \tag{4}$$

With a typical internal resistance value ($R_{\text{int}} \approx 1.0\,\Omega$), the voltage sag under the maximum peak current load of 240 mA (which corresponds to a brief 2C peak burst) is evaluated as:

$$\Delta V = 240 \text{ mA} \times 1.0\,\Omega = 0.240 \text{ V} \tag{5}$$

This limits power rail sag to 240 mV. Because a 240 mV drop is significant, the system layout utilizes a high-value low-ESR bulk tank capacitor network (22 μF ceramic parallel arrays) across the module input rail to absorb the instantaneous pulse energy and prevent local voltage sag from causing premature brown-out triggers.

2.6.2 Regulation Topology

A fully charged LIR2450 cell outputs 4.2 V, which exceeds the maximum safe operating voltage threshold of the module (3.6 V). [cite: 52] Conversely, near its useful discharge cutoff, the cell drops down to 3.0 V. [cite: 52] To bridge this range, a highly stable 3.3 V Low-Dropout (LDO) linear regulator is integrated into the power distribution network. The LDO supports output currents exceeding 500 mA to survive peak Wi-Fi transient surges while maintaining an ultra-low quiescent leakage profile ($I_q < 1\,\mu$A) to maximize power savings during deep sleep.

2.6.3 Empirical Runtime Calculations under Continuous Active Load

To evaluate operational longevity under continuous execution, a profile was modeled with the active system stack (microcontroller cores operating at a moderate clock speed + TIA front-end + Inverter block) drawing a continuous current load of 45 mA:

$$\text{Runtime}_{\text{Theoretical}} = \frac{\text{Battery Rated Capacity (mAh)}}{\text{Average System Current (mA)}} \tag{6}$$

Evaluating this parameter across the 120 mAh LIR2450 configuration yields:

$$\text{Runtime}_{\text{Theoretical}} = \frac{120 \text{ mAh}}{45 \text{ mA}} \approx 2.66 \text{ Hours} \tag{7}$$

Under heavy continuous output operations, the LIR2450 cell safely supports a continuous 45 mA draw (≈ 0.37C continuous rate), providing a reliable, steady operational performance window of approximately 2 hours and 40 minutes on a single charge.

2.7 Module Interface and Pin Assignment

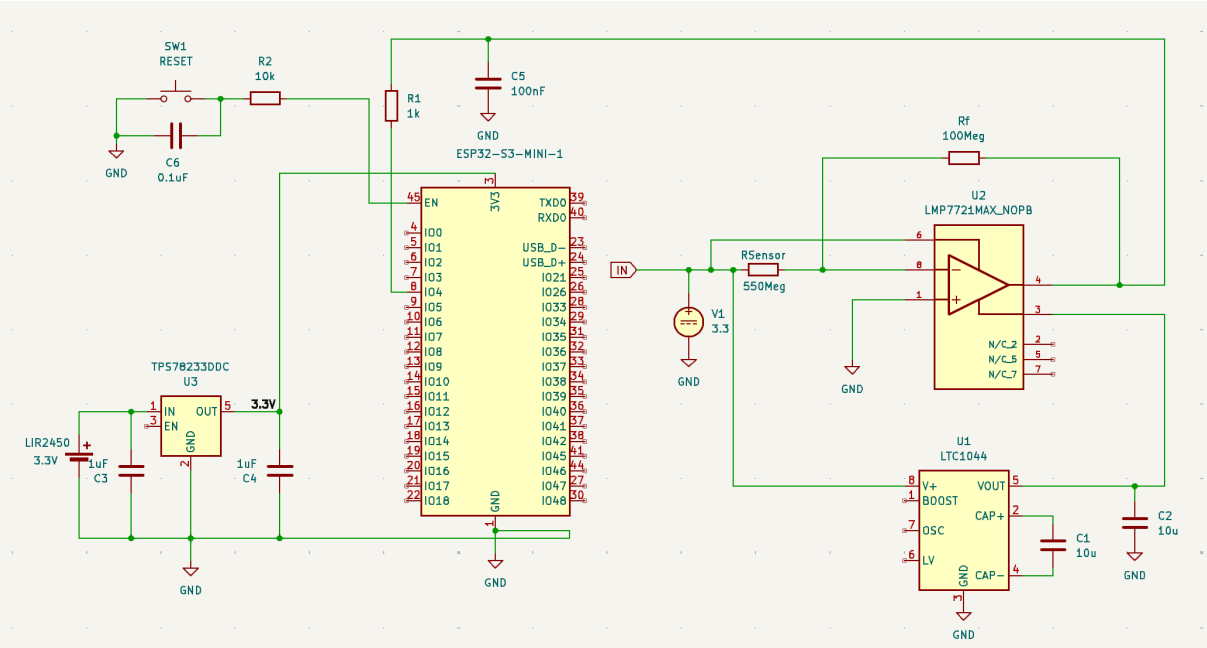


Figure 13: Complete system circuit schematic showing the core interconnected components.

Based on the physical pin layout and reference footprints of the ESP32-S3-Mini-1 module, the hardware connection mapping for the power rails and the analog frontend are established as follows:

- **Regulated Supply (+3.3V Input):** The output from the TPS78233DDC LDO regulator connects directly to **Pin 3 (3V3)** of the module. This rail is locally decoupled with a 100 nF ceramic capacitor (C5) to suppress high-frequency noise.
- **System Board Ground (GND):** System ground is tied directly to the main reference plane at **Pin 1 (GND)**.
- **Analog Summing Junction Monitor Node: Pin 8 (IO4 / ADC1_CH3)** is connected directly to the high-impedance TIA input node (`IN`) to sample the sensor excitation voltage threshold boundary.
- **TIA Output Signal Feedback Line (V_{Mes}):** The output node (**Pin 4**) of the LMP7721 precision transimpedance amplifier is routed to **Pin 5 (IO1 / ADC1_CH0)** to be captured by the internal analog-to-digital converter peripheral.

$$V_{\text{Measured}} = V_{\text{Pin8}} - V_{\text{Pin5}} \tag{8}$$

By executing a differential measurement acquisition cycle across ADC1 Channels 3 and 0, the software architecture quantifies the exact voltage drop across the transimpedance amplifier feedback circuit loop.

Table 5: Hardware Connection Matrix based on Schematic Routing.

Connection Target	**Module Pin**	**Pin Function**	**Internal Peripheral**
Regulated Power (+3.3V)	Pin 3	3V3	Power Input Rail
System Board Ground (GND)	Pin 1	GND	Ground Reference Plane
TIA High-Impedance Input	Pin 8	IO4	ADC1_CH3 (Positive Channel)
TIA Shielded Output (V_{Mes})	Pin 5	IO1	ADC1_CH0 (Negative Channel)

2.7.1 Mathematical Characterization of the Differential Voltage Acquisition

The differential voltage acquisition cycle executed across the analog-to-digital converter channels isolates the meaningful sensor response from the baseline bias potential. Based on the schematic routing configurations, the differential input potential V_{Measured} digitized by the internal peripheral is expressed as:

$$V_{\text{Measured}} = V_{\text{Pin8}} - V_{\text{Pin5}} \tag{9}$$

The DC voltage potential at the high-impedance input monitoring node is explicitly forced by the excitation voltage source V_1:

$$V_{\text{Pin8}} = V_1 = 3.3 \text{ V} \tag{10}$$

The output potential of the LMP7721 precision transimpedance amplifier routed to Pin 5 (V_{Pin5}) is governed by Kirchhoff's Current Law (KCL) at the inverting summing junction. It is a linear function of the input bias potential and the current generated across the functionalized transition metal dichalcogenide nanosheet sensor (I_{Sensor}) scaled by the feedback resistor network (R_f):

$$V_{\text{Pin5}} = V_{\text{Pin8}} - (I_{\text{Sensor}} \cdot R_f) \tag{11}$$

Substituting Equation (3) into Equation (1) yields the direct relationship tracking the voltage conversion parameters:

$$V_{\text{Measured}} = V_{\text{Pin8}} - [V_{\text{Pin8}} - (I_{\text{Sensor}} \cdot R_f)] \tag{12}$$

$$V_{\text{Measured}} = 3.3 \text{ V} - [3.3 \text{ V} - (I_{\text{Sensor}} \cdot R_f)] \tag{13}$$

$$\mathbf{V}_{\text{Measured}} = \mathbf{I}_{\text{Sensor}} \cdot \mathbf{R}_{\mathbf{f}} \tag{14}$$

Therefore, the differential input potential is mathematically decoupled from the absolute +3.3 V baseline operating voltage. It is non-zero during active sampling states and relies entirely on the ohmic properties of the sensor feedback network loop. For instance, given a baseline feedback resistance $R_f = 100$ MΩ, an active trace current steps the measurement response parameters as follows:

- **Quiescent State ($I_{\text{Sensor}} = 0$ A):**

$$V_{\text{Measured}} = 0 \text{ A} \cdot 100 \text{ M}\Omega = 0 \text{ V} \tag{15}$$

- **Active State ($I_{\text{Sensor}} = 10$ nA):**

$$V_{\text{Measured}} = 10 \times 10^{-9} \text{ A} \cdot 100 \times 10^{6} \, \Omega = 1.0 \text{ V} \tag{16}$$

3 PCB Design and Implementation Layout

3.0.1 PCB Design Objectives

The PCB layout was developed with the following objectives:

- Minimize leakage currents around the amplifier input.
- Reduce parasitic capacitance at the summing junction.
- Minimize noise coupling from switching circuitry.
- Protect the high-impedance input node from environmental contamination.
- Improve measurement stability and repeatability.

To achieve these objectives, a guarded PCB architecture was adopted.

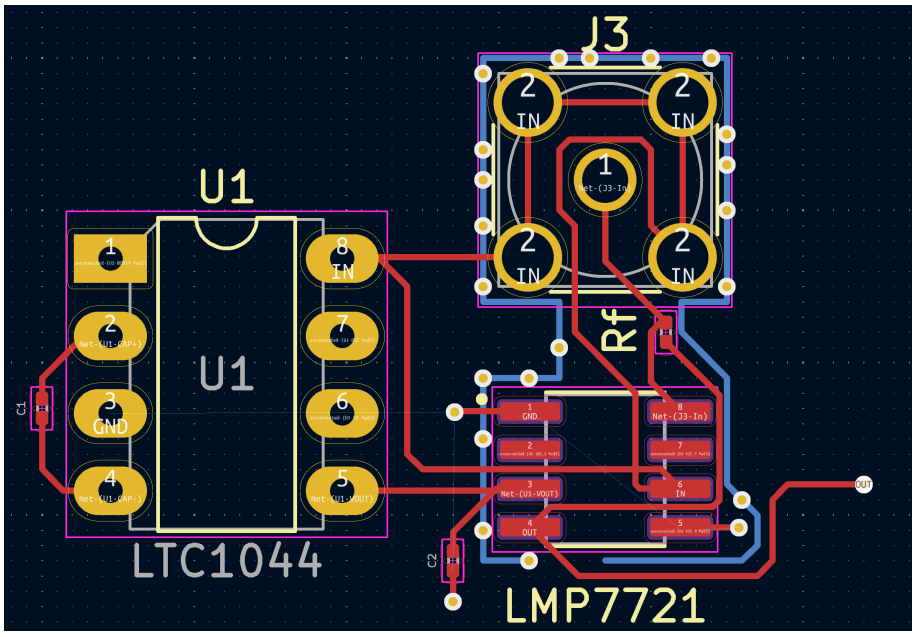


Figure 14: Three-dimensional mechanical trace preview of the high-impedance front-end routing layout.

3.0.2 Component Placement Strategy

The placement of components was optimized to reduce the length of critical high-impedance traces.

The LMP7721 operational amplifier was positioned directly adjacent to the input connector. This arrangement minimizes the exposed area of the sensitive input node and reduces susceptibility to external noise sources.

Similarly, the feedback resistor was placed as close as possible to the amplifier input pins. Keeping the feedback network compact reduces parasitic capacitance and limits leakage paths across the PCB surface.

The LTC1044 charge pump was placed away from the sensitive analog front-end. Since the LTC1044 operates as a switched-capacitor voltage inverter, separating it from the measurement circuitry reduces the possibility of switching noise coupling into the input stage.

3.0.3 High-Impedance Input Region

The most critical portion of the PCB is the high-impedance input region.

This region consists of:

- The center conductor of the input connector (J3),
- The trace connecting the connector to the amplifier,
- The inverting input of the LMP7721,
- The feedback resistor connection.

Because the input current can be extremely small, any leakage current flowing across the PCB surface may significantly affect measurement accuracy.

To minimize these effects, the input trace was made as short as possible and confined within a guarded area.

3.0.4 Guard Ring Design

One of the most important PCB features implemented in this design is the guard ring.

A guard ring is a conductive trace or copper region that surrounds a high-impedance node and is maintained at approximately the same electrical potential. Because the voltage difference between the sensitive node and the guard is very small, surface leakage currents are significantly reduced.

The guard structure surrounds:

- The input connector center conductor,
- The input trace,
- The inverting input pin of the LMP7721,
- The feedback resistor network.

The guard ring effectively intercepts leakage currents that would otherwise flow toward the amplifier input.

This technique is commonly used in electrometers, picoammeters, and precision current measurement instruments.

3.0.5 Top and Bottom Layer Guarding

The PCB was implemented using guard structures on both external copper layers.

3.0.6 Top Layer Guard

The primary guard ring is located on the top copper layer surrounding the entire high-impedance input region.

This guard protects against surface leakage currents generated by moisture, contamination, and residual flux present on the component side of the board.

3.0.7 Bottom Layer Guard

A second guard ring was implemented on the bottom copper layer directly beneath the guarded area.

This additional guard reduces leakage through the opposite PCB surface and provides further isolation from surrounding circuitry.

The combination of top and bottom guard structures forms a three-dimensional protective enclosure around the sensitive input node.

3.0.8 Guard Stitching Vias

The top and bottom guard rings are interconnected using multiple stitching vias distributed around the perimeter of the guarded area.

The purposes of the stitching vias are:

- Electrically connecting the top and bottom guard rings,
- Maintaining a uniform guard potential,
- Improving electrostatic shielding,
- Reducing electric field penetration into the guarded region,
- Minimizing substrate leakage currents.

The stitching vias are placed at regular intervals around the guard perimeter while maintaining sufficient clearance from the high-impedance node.

This arrangement creates a continuous guard structure that surrounds the sensitive circuitry from multiple directions.

3.0.9 Four-Layer PCB Stackup

The design uses a four-layer PCB stackup.

Table 6: PCB Stackup Details

Layer	Function
Layer 1 (F.Cu)	Signal Routing + Guard Ring
Layer 2 (In1.Cu)	Ground Plane
Layer 3 (In2.Cu)	Power Distribution
Layer 4 (B.Cu)	Bottom Guard Ring

The use of dedicated internal planes improves power integrity and reduces electromagnetic interference. However, special precautions were required in the vicinity of the high-impedance input node.

3.1 Microcontroller Core and Power Infrastructure Layout

The digital processing and power regulation node are implemented around an Espressif **ESP32-S3-Mini-1** module (U4) and a Texas Instruments **TPS78233** low-dropout (LDO) linear regulator (U3). This sub-system is optimized to interface a rechargeable LIR2450 Lithium-Ion cell safely to the sensitive dual-core microcontroller architecture while mitigating power rail noise coupling. [cite: 4]

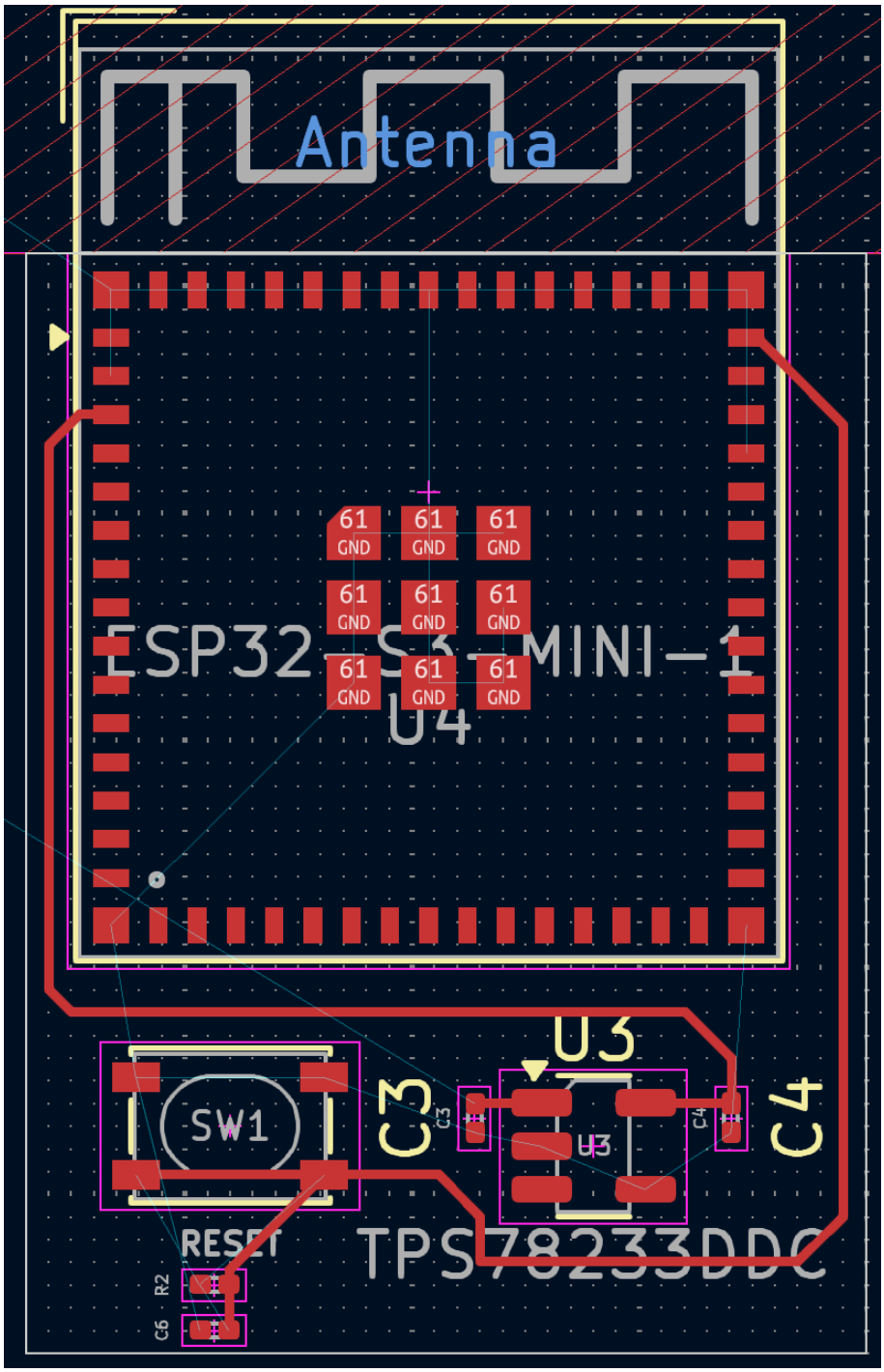


Figure 15: Physical routing layout layout tracing the ESP32-S3-Mini-1 wireless module module and power delivery paths.

3.1.1 Circuit Topology and Schematic Analysis

As illustrated in the system schematic, the power infrastructure bridges the changing terminal voltage of the cell to a stable internal 3.3 V rail:

- **Low-Dropout Regulation:** The TPS78233 (U3) accepts the raw positive battery potential at its input terminal (**Pin 1, IN**) and outputs a fixed 3.3 V rail (**Pin 5, OUT**) dedicated to the microcontroller's main supply channel (**Pin 3, 3V3**).
- **Decoupling Network:** Local filtering configurations are positioned on both sides of the regulation stage. A 1 μF ceramic capacitor (C3) dampens transient source ripples at the regulator input, while an output 1 μF capacitor (C4) provides charge storage to satisfy instant dynamic current loops drawn during radio transitions.
- **Reset Infrastructure:** A physical hardware reset circuit is tied directly to the chip enable pin (**Pin 45, EN**). [cite: 190, 266] The arrangement utilizes a 10 kΩ pull-up resistor (R2) linked to the 3.3 V rail, paired in parallel with a 0.1 μF delay capacitor (C6) and a momentary tactile switch (SW1). This forms an RC delay loop that guarantees the internal power structures reach a stable operational baseline prior to the initialization code execution.

3.1.2 Physical PCB Layout Implementation Analysis

The physical layout properties visible in the PCB design match strict low-noise and high-frequency routing criteria:

Figure 16: Physical routing topology showing the ESP32-S3-Mini-1 module and the associated regulation layout.

- **Antenna Clearance Plane:** The integrated PCB antenna array at the top region of U4 sits completely isolated from any routing layers, ground polygons, or signal vias. [cite: 6, 174] This clear boundary space guarantees proper characteristic impedance values and prevents signal attenuation over the 2.4 GHz wave path.
- **Thermal Pad Ground Array:** The physical center cavity of the module footprint utilizes a centralized matrix of grid arrays connected directly to the primary reference ground plane. [cite: 208, 245] This low-impedance contact point functions as both an electrical common path and a localized heat sink to dissipate thermal loads generated by the processor core.
- **Trace Routing Strategy:** Power delivery channels are routed using thick trace widths to minimize parasitic series resistance and local voltage drops. The critical 3.3 V distribution line leaves capacitor C4 and traces along a dedicated path to loop cleanly into the module supply pad, minimizing loop areas that might cause electromagnetic susceptibility.

4 Conclusion

This report detailed the successful hardware architecture design, functional verification, and physical layout optimization of the embedded wireless acquisition node for the **SENS-CO** wearable gas sensing project. By addressing the strict low-power and high-impedance constraints required to measure sub-nanoampere currents from functionalized transition metal dichalcogenide nanosheets, a robust system topology was achieved.

The analog front-end parametric evaluation confirmed that while the ADA4530-1 offers unrivaled high-impedance isolation, the LMP7721 precision amplifier provides the rapid transient settling behavior necessary for duty-cycled battery applications. Combined with the LTC1044 switched-capacitor voltage inverter, the system establishes a reliable dual-supply network capable of processing bipolar signals seamlessly from a single LIR2450 Lithium-Ion cell. Furthermore, the firmware-assisted differential ADC strategy isolates the sensor response while protecting the ESP32-S3 microcontroller core from harmful negative potentials.

On the physical design level, the implemented four-layer stackup incorporates rigorous three-dimensional guard structures, including top and bottom guard rings stitched with low-impedance vias. These layouts effectively isolate the ultra-low current sensor traces from environmental leakage currents and the high-frequency switching noise of the power and RF circuits. Ultimately, this complete hardware platform fulfills the strict criteria of portability, autonomy, and measurement integrity required for modern wearable gas monitoring applications.

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